SAGAR KUMAR

Software Engineer | C++ Developer | Embedded Systems | Industrial Communication Protocols

Chennai, India | +91-9755005359 | sarkarsagar20100@gmail.com | LinkedIn: sagar-kumar-b23771205

PROFESSIONAL SUMMARY

Software Engineer with 3.5+ years of experience in C++ development for embedded systems, industrial communication protocols, and drive firmware. Experienced in BACnet MSTP/IP protocol development, object-oriented design, algorithms, hardware integration, debugging, unit testing, requirements engineering, and the full software development lifecycle. Hands-on experience creating PDS, HLD, LLD, CRS, and PRS documentation for BACnet MSTP/IP features, from requirements through implementation, verification, traceability, and release.

TECHNICAL SKILLS

Programming: C++, C++11, C++14, C++17, C, Python, SQL, JSON

C++ / Software Engineering: Object-Oriented Programming (OOP), STL, Data Structures & Algorithms (DSA), Design Patterns, Memory Management, Pointers, References, Classes, Inheritance, Polymorphism, Templates, Exception Handling, Multithreading, Debugging, Unit Testing, Integration Testing

Protocols: BACnet MSTP, BACnet/IP, Modbus, CIP, EtherNet/IP

Design & Requirements: HLD, LLD, PDS, CRS, PRS, Architecture Diagrams, Requirements Gathering, Traceability, Verification Records, Release Planning

Testing & Debugging: Google Test, Unit Testing, Integration Testing, Interoperability Testing, BACnet Compliance Testing, Wireshark, Hardware Debugging

Tools & ALM: Git, GitHub, Azure DevOps, Siemens Polarion, Enterprise Architect, Draw.io, YABE, VTS, PuTTY, Xilinx Vivado, Xilinx SDK, MyDrive Insight, Wago Configurator

PROFESSIONAL EXPERIENCE

Danfoss Industries Pvt. Ltd. — Chennai, India

Software Engineer | Feb 2023 – Present

BACnet MSTP/IP Protocol Development — Danfoss iC7 Series Drive Firmware

BACNET MSTP/IP — C++ FIRMWARE DEVELOPMENT

- Architected and implemented BACnet event handling and Change of State (COS) algorithms, enabling real-time property monitoring, event generation, and subscriber notification in compliance with ASHRAE 135.
- Designed and integrated BACnet Intrinsic Reporting with event state machines, supporting error, offnormal, and normal transitions with configurable thresholds and time delays.
- Implemented BACnet Device Object, Network Port Object (MSTP), Binary Value Objects, and Trend Log Object with persistent buffer management.
- Integrated the BACnet MSTP license feature into drive firmware, enforcing controlled feature activation and licensing compliance at the firmware layer.
- Built dynamic storage mechanisms for BACnet object properties, supporting data persistence across warm and cold restarts and improving field reliability.

PRODUCT DOCUMENTATION, ARCHITECTURE & REQUIREMENTS

- Created and maintained full Product Data Specification (PDS), High-Level Design (HLD), Low-Level Design (LLD), Customer Requirement Specification (CRS), and Product Requirement Specification (PRS) documentation for BACnet MSTP/IP features and protocol modules.
- Produced HLD and LLD architecture documentation and diagrams using Draw.io and Enterprise Architect for firmware and communication-protocol modules.
- Managed end-to-end requirements in Siemens Polarion, including requirements gathering, bidirectional traceability, verification records, and release planning.

TESTING, DEBUGGING & HARDWARE INTEGRATION

- Developed and executed unit tests for BACnet protocol modules and CIP Security certificate management using Google Test, providing protocol conformance and regression coverage.
- Performed firmware flashing on control cards using Xilinx Vivado and hardware-level debugging using Xilinx SDK to isolate and resolve embedded-system faults.
- Conducted deep-packet inspection and protocol troubleshooting for BACnet, Modbus, and CIP traffic using Wireshark, validating communication against specifications.
- Performed manual BACnet protocol validation and interoperability testing using YABE for functional compliance across vendor devices.
- Managed real-time firmware variable inspection and patching through PuTTY serial console for live system verification during development and field debugging.

CRITICAL PCR FIXES & PRODUCT RELEASES

- Diagnosed and resolved critical PCRs (Problem/Change Reports) against iC7 drive general-release builds, ensuring firmware stability and compliance for HVAC market deployments.
- Contributed to multiple product release cycles by implementing new features alongside targeted defect fixes while maintaining backward compatibility with existing BACnet protocol behavior.
- Collaborated with cross-functional teams to prioritize and resolve high-severity issues within Agile sprint cycles, reducing field-reported defects in production firmware.

DANFOSS AI/ML INITIATIVE

- Created an MCP (Model Context Protocol) solution to parse and retrieve information from local documents and SharePoint, and integrated it with Visual Studio 2026 and Visual Studio Code for developer-accessible document intelligence.
- Created a Siemens Polarion MCP to gather and retrieve software requirements directly inside Visual Studio 2026 and Visual Studio Code, improving access to requirements during development.

EDUCATION

Veltech R&D Institute of Technology and Science, Veltech University — Bachelor of Technology, Computer Science and Engineering | CGPA: 7.48/10 | 2019 – 2023

Sector-10 CBSE — High School, Computer Science & IT, First Division | 2019

CERTIFICATIONS

- Palo Alto Networks Cybersecurity — 2021
- AWS Cloud Foundations — Digital Badge — 2021
- NVIDIA Fundamentals of Deep Learning — 2020

LANGUAGES

English (Professional Working) | Hindi (Professional Working) | Bengali (Limited Working)

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